

Abhay Rathor

Python Developer | GenAI & RAG Applications

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PROFESSIONAL SUMMARY

Python Developer with 3 years of experience building AI-powered applications and scalable backend systems using Python, Django, LangChain, and Retrieval-Augmented Generation (RAG). Experienced in integrating LLM-powered solutions, semantic search, vector databases, and conversational AI to build intelligent applications. Skilled in developing secure REST APIs with Django REST Framework, JWT authentication, Celery, Redis, and MySQL/PostgreSQL. Proficient in React.js for frontend development and deploying production-ready applications using Docker, Kubernetes, AWS, Terraform, CI/CD pipelines, and Linux environments. Hands-on experience developing industrial automation solutions, real-time PLC monitoring systems, and AI-driven troubleshooting platforms.

WORK EXPERIENCE

Python Developer — Technoviz Automation Solutions, Gurgaon

June 2023 – Present

- Engineered and maintained scalable full-stack web applications using Django, React.js, and Python, improving backend response efficiency through database query optimization and efficient API design.
- Designed and implemented RESTful APIs enabling secure, scalable communication between frontend and backend systems.
- Implemented JWT-based authentication and role-based access control, strengthening application security and ensuring secure user access management.
- Automated background job processing using Celery with Redis, enabling real-time data synchronization and reducing manual operational effort.
- Optimized MySQL and PostgreSQL database schemas and queries, improving data integrity and application performance.
- Established CI/CD pipelines using Docker and GitHub Actions, streamlining deployment workflows and accelerating software releases.
- Managed Linux-based application deployments on AWS, ensuring high availability, scalability, and reliable production environments.
- Collaborated with cross-functional teams using Git and GitHub, maintaining clean version control practices and participating in code reviews.

PROJECTS

Maintenance Mate – AI-Powered Industrial Motor Diagnosis Assistant

Stack: Python · Django · LangChain · Gemini API · ChromaDB · RAG · Celery · Redis · Docker

- Developed an AI-powered assistant to diagnose industrial motor and relay faults by understanding user queries and retrieving relevant technical information using Retrieval-Augmented Generation (RAG).
- Integrated LangChain with the Gemini API to provide context-aware troubleshooting, maintenance guidance, and fault resolution recommendations.
- Built a semantic knowledge base using ChromaDB and vector embeddings, enabling intelligent search across motor manuals, relay documentation, and historical fault records.
- Implemented conversational memory to maintain context across multiple user interactions, delivering more accurate and personalized responses.
- Automated email notifications to share fault reports, troubleshooting recommendations, and the latest equipment information with users.

- Used Celery and Redis for asynchronous document processing, email delivery, and background tasks, while deploying the application using Docker.

Inspection & Audit Management System

Stack: Django · React.js · MySQL · JWT · Celery · Redis · Docker · CI/CD

- Architected a role-based web application supporting multi-level inspection and audit workflows for 50+ users.
- Implemented JWT authentication, dynamic checksheets, and automated CSV/PDF report generation, eliminating manual reporting effort.
- Integrated Celery and Redis for asynchronous task processing, enabling real-time status updates without blocking the main thread.
- Containerized the application with Docker and configured CI/CD pipelines, reducing deployment time and improving release reliability.

OEE Monitoring & Control Device Software

Stack: Python · Django · Celery · Redis · Docker · AWS · Linux

- Developed a real-time PLC-integrated system to monitor machine availability, performance, and quality metrics for industrial equipment.
- Implemented automated fault tracking and OEE metric dashboards, enabling proactive maintenance decisions.
- Used Celery and Redis for real-time data synchronization across multiple devices and background job processing.
- Deployed on Linux systems with Docker and basic AWS setup, ensuring reliable 24/7 uptime in a production environment.

SKILLS

AI & LLM:	LangChain, Retrieval-Augmented Generation (RAG), ChromaDB, Vector Embeddings, Semantic Search, Prompt Engineering, Gemini API
Backend & API Development:	Python, Django, Django REST Framework, Flask, REST API, JWT, Celery, Redis
Frontend Technologies:	React.js, JavaScript (ES6+), HTML5, CSS3
Databases:	MySQL, PostgreSQL, SQL
Data & Libraries:	NumPy, Pandas
DevOps & Cloud:	Docker, Kubernetes, AWS (EC2, S3), Terraform, CI/CD, Linux
Version Control:	Git, GitHub

EDUCATION

Master of Computer Applications (MCA) — Uttaranchal University (2026 – 2028, Pursuing)

Bachelor of Computer Application — Maharshi Dayanand University, Rohtak (Completed 2026)

CERTIFICATIONS

- Advanced Python Programming
- MySQL Database Management
- DevOps Fundamentals & Tools – Docker, Kubernetes, AWS, Terraform, CI/CD